

Multiple Choice

1. **Answer:** A

Options: A) 5×10^5 nt; B) 1×10^5 nt; C) 7×10^5 nt; D) 3×10^5 nt; E) 6×10^5 nt

Note: Correct option: A - 5×10^5 nt

2. **Answer:** C

Options: A) lower order to higher order; B) chaos to equilibrium; C) higher order to lower order; D) disorder to organization; E) equilibrium to higher order

Note: Correct option: C - higher order to lower order

3. **Answer:** B

Options: A) changes its atomic number but remains the same element.; B) becomes an atom of a different element always.; C) remains the same element and isotope.; D) becomes a different element with the same atomic mass.; E) becomes an atom of a different element sometimes.

Note: Correct option: B - becomes an atom of a different element always.

4. **Answer:** B

Options: A) 4 mm; B) .833 mm; C) 1 mm; D) 1.5 mm; E) 0.5 mm

Note: Correct option: B - .833 mm

5. **Answer:** C

Options: A) 2.5 hp; B) 2.7 hp; C) 1.8 hp; D) 1.2 hp; E) 3.0 hp

Note: Correct option: C - 1.8 hp

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '0.01 nt- m'.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** True

Options: A) True; B) False

Note: Matches the MMLU-Pro answer key.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** Initial velocity: 10 ft/s, Time of motion: 20 s. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 19.9 w/m^2 . Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key